Daniel Liu

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Objective

Graduated Nuclear Engineer with a passion of working on big projects and an interest in data science / simulations.

Education

University of Ontario Institute of Technology Oshawa, ON

Bachelor of Engineering in Nuclear Engineering, Co-op *2013 – 2018*

Minors: Computer Science & Chemical Engineering

Thesis: Improving procedural efficiency, and emergency response for nuclear & energy workers using Augmented Reality. Supervised & Funded by UOIT and OPG.

Summary of Qualifications

- Effective communication, commercial awareness and leadership skills. Experienced on working with federally funded Research & Development projects in the nuclear sector and securing $50,000 towards the development augmented reality tools for workers.
- Intermediate knowledge of data science and analytics skills demonstrated through the practice of various problems using machine learning packages, Juypter for reporting, and MCNP for radiation shielding simulations
- Effective communication, commercial awareness and leadership skills. Developed a Proof-of-Concept iOS application to win a $5,000 innovation prize toward a mobile game to help millennial's invest in entertaining manner

Experience

Ontario Power Generation September 2018 – Present

Refurbishment Project Engineer, Darlington Nuclear

- Acting as a Project Engineer/Leader for a $0.5 B nuclear project to coordinate milestones from Engineering, to Procurement and Construction with multiple contractors to successfully commission a facility under extremely tight deadlines
- Using Oracle Primavera P6 to schedule thousands of contractor duties and objectives. Collaborating with multiple engineers and managers to successfully execute work under multiple constraints while avoiding unnecessary work

Canadian Nuclear Laboratories May 2018 – September 2018

Research Officer / Engineer, Chemical Engineering R&D

- Provided detailed project updates to federally run Atomic Energy of Canada Limited (AECL) to maintain funding of department Science & Technology experiments worth >$ 200 k for further development and eventual commercialization
- Provided programming and engineering skills to develop and scale-up an experimental process to separate Tritium, a radioactive isotope of Hydrogen. Successfully proposed the viability of this technology for increased investment to supply the future demand of Tritium for fusion energy

Ontario Power Generation May 2016 – September 2017

Performance Engineering Intern, Tritium Removal Facility & Heavy Water Management Building

- Executed SQL queries on big data sets to determine common failure rates of in-field equipment. Used this data to analyze, organize, and report optimal maintenance routines for critical nuclear equipment.
- Developed a robust Excel VBA program to simulate nuclear reactor radioactivity build-up using nuclear reactor physics algorithms to organize decontamination schedules

Projects and Activities

McKinsey & Company Hackathon: Analyzed if a requested loan is approved or not, based on the features of the applicant (e.g., income, gender, city), and that of the loan (e.g., loan amounts, interest). Used XGBoost for analysis.

Scotiabank Hackathon: Delivered an interactive and entertaining mobile game to help reduce user debt via in-app purchases. Participated with eight software engineers and graphics designers to win a $5,000 prize.

Programming/Technologies

Proficient: SolidWorks, MATLAB, VBA, HTML & CSS, Oracle Primavera P6, MS Office, C/C++, Java, LaTeX

Used: Javascript, GitHub, Node.js, Python, SAP Ariba, MySQL

Touched: ANSYS, Visual Studio, Android Studio, Fortran, Tableau, RStudio, TensorFlow, Keras, DMLC XGBoost